

Decision Making Under Uncertainty: 95-760

Homework 3

Due: Monday, September 28th, 2026 at 11:59 pm

Please submit a PDF to Gradescope and the Excel file to Canvas by the submission date. Make sure you indicate the page for each question when you submit it to Gradescope. If you do not select the pages for the questions, your question will not be graded. Write your name(s) on the first page of the files you submit.

You can handwrite the homework and scan it to upload it to Gradescope. If you hand write please make sure that your work is organized. Many of the problems will require you to construct a spreadsheet. In these cases, you should copy a screenshot of the spreadsheet into the PDF that you submit, and also clearly document the formulas that you used. The PDF submitted should be self contained. In other words, include all necessary screenshots and documentation for the graders to understand your answer. To help the grader, excel spreadsheets should have all formula and solver settings documented. You can find an example on Canvas.

If you need to draw a picture, then either scan or take a picture of it, and copy it into the PDF that you submit. You should also submit the Excel document you use to solve the problems. You should use the same Excel file for all problems in the Homework. Create a new tab for each problem.

You can submit individually or form groups of up to 3 students and submit one copy as a group. Make sure to select all team members on Gradescope. In the PDF file, please mention the person who will upload the Excel file to Canvas. Only one person needs to upload the Excel file to Canvas.

PROBLEM 1

A new company wants to enter the USA market. The company produces and sells electric cars. They want to target 5 main cities; Pittsburgh, Chicago, New York, Washington D.C., and Boston. Based on some market research, they expect to sell 12,000 cars in their first year of operations, distributed as follows: 1,000 cars in Pittsburgh, 3,000 in Chicago, 3,000 in New York, 1,000 in Washington D.C. and 4,000 in Boston. The company has 4 possible options for manufacturing facilities. Each of these has a construction cost, production cost (per car), minimum production if facility is built, and maximum capacity. The table below shows the characteristic of each potential facility:

Facility	Construction cost	Production cost (per unit)	Minimum Production	Maximum Capacity
Plant 1	\$2,100,000	\$380	2,000	5,000
Plant 2	\$2,800,000	\$165	3,000	7,000
Plant 3	\$2,400,000	\$155	4,000	8,000
Plant 4	\$2,700,000	\$170	2,500	6,000

Let's first assume that the company wants to minimize construction cost while making sure that there is enough capacity to serve the demand. We do not need to consider production cost or transportation cost to each city.

- Formulate the problem as an integer programming model. Clearly define your decision variables, constraints, and objective function.
- Solve the problem using Excel Solver.
- Present the results to the company.

We now account for production costs, but not transportation costs yet. We want to determine the set of facilities to build and the number of electric vehicles to produce in each facility, to minimize the sum of construction costs and production costs, while ensuring that enough electric vehicles will be produced across all the facilities to serve the total customer demand.

- d) Formulate the problem as an integer programming model. Clearly define your decision variables, constraints, and objective function.
- e) Solve the problem using Excel Solver.
- f) Present the results to the company.

Now, we consider all costs. In other words, we need to decide which facilities to build, how many electric vehicles to produce in each plant, and how many electric vehicles to transport from each plant to each city in order to minimize the sum of facility construction costs, production costs, and transportation costs. The cost per car between each facility and city is shown in the table below:

	Pittsburgh	Chicago	New York	Washington, D.C.	Boston
Plant 1	\$18	\$35	\$50	\$30	\$20
Plant 2	\$15	\$10	\$12	\$10	\$12
Plant 3	\$40	\$50	\$30	\$35	\$25
Plant 4	\$32	\$25	\$20	\$25	\$40

- g) Formulate the problem as an integer programming model. Clearly define your decision variables, constraints, and objective function.
- h) Solve the problem using Excel Solver.
- i) Present the results to the company.

PROBLEM 2

The 10th edition of the FIFA Women's World Cup is scheduled for next summer in Brazil. The tournament opens with a group stage consisting of eight groups of four teams each, with the top two teams progressing from each group to a knockout tournament starting with a round of 16 teams. Planning the World Cup brings many logistical challenges, including deciding when and where each team plays, how to transport the athletes, and allocating staff and equipment, among others. Your task is to help FIFA address some of these challenges.

For this problem you will focus into one of the eight groups. The group consists of four countries: Colombia, New Zealand, Germany, and Philippines. Each team plays three matches, one against each other team. This means that there are 6 total matches (shown below).

Match	Teams
1	Colombia–Germany (C-G)
2	Colombia–Philippines (C-P)
3	New Zealand–Germany (NZ-G)
4	New Zealand–Philippines (NZ-P)
5	Colombia–New Zealand (C-NZ)
6	Germany–Philippines (G-P)

FIFA needs help deciding on the day of the matches for each of the teams. In addition to the information above, you know that:

- The matches occur in a time frame of 14 days
- For fairness, the final two matches should be played at the exact same time on day 14. This prevents teams from gaining an unfair tactical advantage by knowing the other match's outcome.
- There can not be more than 2 games on a day

- All matches should be played on the 14 days
- A team can play at most one match every day

Your objective is to create the match schedule, in other words, decide which day will each team play with the objective of maximizing visibility score. The table below indicates the visibility score for each country and day.

Day	C-G	C-P	NZ-G	NZ-P	C-NZ	G-P
1	185	170	183	168	178	175
2	130	125	125	120	125	125
3	110	120	115	125	115	120
4	138	140	138	140	140	140
5	175	170	182	177	177	175
6	120	125	125	130	125	125
7	110	110	110	110	110	110
8	144	142	142	140	142	142
9	192	183	177	168	175	185
10	130	130	130	130	130	130
11	120	120	120	120	120	120
12	156	153	153	150	153	153
13	180	175	182	175	188	181
14	150	150	150	150	150	150

- Formulate the problem as an integer programming model. Clearly define your decision variables, constraints, and objective function.
- Solve the problem using Excel Solver.
- Present the results to FIFA.
- Looking at the solution, FIFA realized that teams are playing on consecutive days. They let you know that team needs at least 2 days of rest between matches to guarantee optimal performance and rest. That is, if the team plays on day 1 they can play again in day 4 or after. Mathematically, formulate this constraint. Hint: you can multiply variables by the index. For example if you have $x_{i,j} \geq 2$, you can have $i \cdot x_{i,j} \geq 2$